Machine Learning Project Report

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1. Introduction

In this project, I learned how machine learning models can classify and predict data. The main idea of this project is to use machine learning and artificial intelligence techniques to predict patient health deterioration using medical data such as blood sugar levels and blood pressure. The project aims to help hospitals monitor patients more accurately and detect dangerous health changes at an early stage. Different machine learning models were trained and compared to determine which model provides the best prediction performance. I also wanted to understand how neural networks work compared to traditional machine learning models.

2. Dataset Description

The dataset used in this project was divided into training and testing data. It contained several features that helped the models make predictions. The data was prepared and cleaned before training so the models could learn correctly and improve prediction accuracy. The purpose of the dataset was to evaluate how well different machine learning models can classify data and make accurate predictions related to patient health conditions.

3. Models Used

In this project, I used three machine learning models: Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN). The results showed that KNN achieved the best performance compared to the other traditional models. Random Forest also provided strong and stable results, while Logistic Regression achieved good but slightly lower performance.

4. Neural Network

I also trained a neural network model using TensorFlow/Keras. The model was evaluated using the same metrics as the other models. The neural network achieved acceptable accuracy, but the Precision and Recall values were very low. This means the model failed to detect many positive cases, making its overall performance weaker than the other models.

5. Results and Comparison

After comparing all the models, KNN achieved the best overall performance because it had the highest Accuracy and F1-score. Random Forest also produced strong and stable results. However, the neural network performed worse than the other models. Although its Accuracy was above 92%, the Recall and Precision values were extremely low. This demonstrates that high Accuracy alone does not always indicate that a model performs well.

6. What I Learned

From this project, I learned how to train and compare machine learning models using Python libraries such as Scikit-learn and TensorFlow. I also learned how to evaluate models using Accuracy, Precision, Recall, and F1-score. One important lesson I learned is that some simple machine learning models can sometimes perform better than neural networks depending on the dataset and the problem being solved.